

Assignment nr. 1 – Transmission Lines Scattering matrix

Assigned: Wednesday March 04, 2026
Due: Wednesday March 18, 2026 at class starting

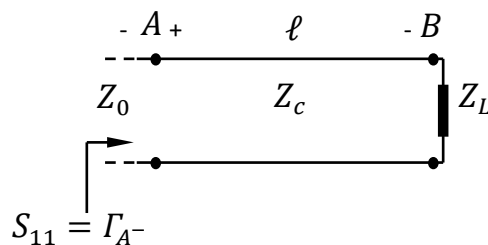
Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to solve **individually** the exercises;
- to submit the solution **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

Problem nr. 1

- a) Design a lossless Quarter Wave Transformer (QWT), consisting of a transmission line section with length $\ell = \lambda_0/4$ at the (design) center-band frequency f_0 and characteristic impedance $Z_C = \sqrt{Z_0 Z_L}$, that matches a load with impedance Z_L to a specified input Z_0 .



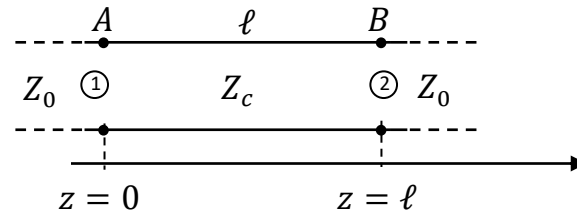
In particular, evaluate Z_C , for the following two cases:

1. $Z_0 = 50 \Omega$, $Z_L = 100 \Omega$;
2. $Z_0 = 50 \Omega$, $Z_L = 200 \Omega$.

- b) Using Matlab™, write a script to compute the frequency behavior of $|S_{11}(f)|$ set in the range $[0, 2f_0]$, and plot it as a function of f/f_0 , for the two different cases.

Problem nr. 2

Consider the lossless transmission line (TL) below, with length $\ell = \lambda_0/2$ at the frequency f_0 and characteristic impedance Z_C . The line can be considered as a 2-port network, and therefore it can be characterized by the scattering matrix $[S]$, computed with respect to the reference impedances Z_0 at both ports.



Using Matlab™, write a script to compute the frequency behavior of $|S_{21}(f)|$ set in the range $[0.5f_0, 1.5f_0]$, and plot it (in dB) as a function of f/f_0 , for the following cases:

1. $Z_0 = 50 \, \Omega, Z_c = 100 \, \Omega$;
2. $Z_0 = 50 \, \Omega, Z_c = 150 \, \Omega$;
3. $Z_0 = 75 \, \Omega, Z_c = 10 \, \Omega$.